

Project Backlog

Bareturn0; - Team 16

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Problem Statement:

Modern educational methods often lack engaging and interactive ways to teach programming concepts and foundational skills like math. Many students struggle with abstract topics such as loops, pointers, and conditional statements when learning programming for the first time. Traditional teaching approaches can feel unrelatable, failing to bridge the gap between theory and practical application. This project aims to address these challenges by creating an educational card game that incorporates interactive learning, gamification, and storytelling to make learning programming and basic math fun, engaging, and memorable.

Background Information:

- Audience:

Team 16's project is designed for a diverse audience, including students and beginners in programming who want to strengthen their understanding of basic coding concepts and math in an engaging way. By gamifying the learning experience, the project appeals to those who prefer interactive and hands-on learning environments. Additionally, it targets educators who are looking for innovative tools to make teaching programming more approachable and effective. Casual learners with a general interest in coding and problem-solving will also find this project accessible and rewarding.

- Similar Platform:

The project takes inspiration from existing platforms like **CodeCombat** and **Slay the Spire**, combining elements of educational gaming with strategic card-based mechanics. Unlike platforms that focus solely on programming challenges or traditional game mechanics, Team 16's project offers a hybrid system where learning concepts like loops, conditionals, and pointers are integrated into a fun and narrative-driven card game. This unique blend of education and entertainment makes the project stand out as a novel solution in both the gaming and learning spaces.

- Limitations:

While the project offers an engaging way to learn programming and math basics, there are some limitations. It primarily focuses on beginner-level concepts and may not meet the needs of advanced learners. The game's initial release may also be platform-specific, potentially limiting accessibility for some users. Additionally, as a self-paced game, it may not provide real-time guidance for users struggling with specific topics. However, these limitations present opportunities for future updates and expansions, ensuring the project continues to evolve and cater to a wider audience.

Requirements:

Functional Requirements

- Welcome and Main Menu:
 1. As a user, I would like to see a loading animation when the game starts, so that the game feels engaging and polished.
 2. As a user, I want the option to skip the loading animation and directly access the main menu, so I can save time when re-entering the game.
 3. As a user, I would like to see a main page after loading the welcome animation, including the function buttons I need(log-in, settings, etc.).
 4. As a user, I would like to create my own account.
 5. As a user, I would like to log in and log out of my own account.
 6. As a user, I want to access settings to adjust volume and sound effects.
 7. As a user, I would like to view statistical data(percentage) on game progress, such as card collections and features collection progress.
- Save Management:
 8. As a user, I would like to create multiple save files under one account, so I can have different game progressions.
 9. As a user, I would like to load a specific save file from the list of my saves.
 10. As a user, I would like to delete save files that I no longer need.
- Achievements
 11. As a user, I would like to unlock achievements with dates of receiving for significant milestones in the game, so I feel a sense of progression and accomplishment.
 12. As a user, I want an on-screen notification when I unlock an achievement, so I feel immediate satisfaction and recognition.
 13. As a user, I would like to earn rewards (e.g., in-game currency, items, or cosmetics) for completing achievements, so I feel incentivized to pursue them.
 14. As a user, I want a dedicated page in the game to display all the achievements I have unlocked, so I can revisit and showcase my accomplishments.
- Game Background and Core Flow
 15. As a user, I would like the option to pause the game and return to the main menu, without saving the progress.
 16. As a user, I want to see an option to restart the current level from the pause menu, so I can quickly try again if I make a mistake.
 17. As a user, I want to see a detailed breakdown of rewards (e.g., experience points, loot) on the victory settlement screen, so I know what I've earned.
 18. As a user, I would like the failure screen to provide suggestions or tips for improving in the next attempt, so I feel motivated to try again.

19. As a user, I want the victory and failure screens to offer quick options to replay the level or return to the map screen, so I can easily decide my next step.
20. As a user, I would like a narrative story to provide background for the game world and context for my actions.
- User Page
 21. As a user, I would like to view my profile information in the user page.
 22. As a user, I want the ability to modify my personal details (e.g., username, email, and password) from the user page, so I can keep my account up to date.
 23. As a user, I want to upload and update a custom profile picture, so I can personalize my account.
 24. As a user, I want a quick link to contact support or provide feedback directly from my user page, so I can resolve issues or share suggestions conveniently.
- Card Game Mechanics
 25. As a user, I would like to preview card attributes to understand their effects before using them.
 26. As a user, I would like to select the functional cards I need before entering battle rounds to customize my strategy.
 27. As a user, I would like to filter and sort cards by attributes (e.g., attack, defense, or special effects) when selecting my functional cards before a battle, so I can quickly find the cards I need.
 28. As a user, I would like to save multiple custom decks for different battle scenarios, so I can switch strategies easily.
 29. As a user, I want both my character and the boss have health points, so the game has a clear win or lose condition based on health depletion.
 30. As a user, I want each card I choose to reduce my stamina by a specified amount, so I need to carefully manage my stamina during each turn.
 31. As a user, I want my stamina to regenerate at the start of each round, so I can have a fresh pool of resources to play new cards.
 32. As a user, I want the game to operate in a turn-based format, so I have time to plan my strategy and respond to the boss's moves.
 33. As a user, I want each turn of the battle be under a time count-down mode.
 34. As a user, I want to get new cards at the beginning of each round, so I have a fresh set of options for every turn.
 35. As a user, I want discarded cards to go into a discard pile, so my deck remains organized and cards are not reused immediately.
 36. As a user, I want the discard pile to be reshuffled into the draw pile when the draw pile is empty, so I can continue to play without interruption.

- 37. As a user, I want the boss to try to defeat me by playing cards strategically, so that the combat feels fair, challenging, and engaging.
- 38. As a user, I want the boss to have unique abilities or cards that challenge my strategy, so the battles feel varied and exciting.
- 39. As a user, I want to see a visual indication of changes in health, stamina, and card piles during the game, so I can easily track the current state of the battle.
- Map and Exploration
 - 40. As a user, I want to unlock new map areas by either completing specific levels or learning certain skills, so I can explore the game world in a way that aligns with my progress and choices.
 - 41. As a user, I want to see a figure of mine on the map to see which level I'm currently on.
 - 42. As a user, I want the map to restrict access to incomplete levels while allowing me to freely revisit previously completed levels, so I can replay them for practice or better rewards.
 - 43. As a user, I want to include optional levels in the map which focus on specific skills or learning topics, so I can choose to strengthen areas that interest me.
 - 44. As a user, I want to zoom in and out of the map with ease, so I can navigate between detailed sections and the overall layout effortlessly.
 - 45. As a user, I want to unlock map-themed achievements, so I feel motivated to explore thoroughly.
- Reward System
 - 46. As a user, I would like to earn rewards for completing levels and challenges, which can be used to exchange for blank functional cards or other upgrades.
 - 47. As a user, I would like to receive different types of rewards (e.g., blank cards, rare items, in-game currency, or exclusive abilities), so my gameplay feels more varied and engaging.
 - 48. As a user, I would like bonus rewards for completing levels with certain conditions (e.g., within a time limit or without losing health), so I can aim for more ambitious goals.
 - 49. As a user, I would like the rewards for challenges to scale with difficulty, so I feel incentivized to take on harder challenges.
- Learning Features
 - Learning printf:
 - 50. As a user, I would like to name my character while learning the printf function, so I can understand how to format strings and personalize my game experience.
 - 51. As a user, I want the game to have NPC interactions to teach me the basics of the printf function, so I can use them effectively.
 - 52. As a user, I would like mini-games to practice printf with customized exercises.

- 53. As a user, I want to earn my first achievement, "Person You Know Who," by demonstrating mastery of the printf function, so I feel motivated to continue learning.
- 54. As a user, I want NPCs to offer increasingly complex printf challenges (e.g., combining text with numbers or formatting alignment), so I can gradually improve my skills.
 - Learning Basic Math:
- 55. As a user, I want a blacksmith NPC to teach me basic math operations (addition, subtraction, multiplication, division) through interactive examples, so I can learn while engaging with the story.
- 56. As a user, I would like to complete an independent quest helping a NPC manage their finances, where I apply math skills to solve real-world-like problems.
- 57. As a user, I want to unlock the in-game shop and earn the achievement "Live for your own" after completing a series of math-based challenges, so I feel a sense of accomplishment.
- 58. As a user, I want the math challenges to scale in difficulty, starting with simple calculations and progressing to multi-step problems, so I can gradually improve my skills.
 - Learning if-else and for/while Loops:
- 59. As a user, I want to learn conditional statements and loops through a maze game where NPCs provide guided examples, so I can practice these concepts in a visual and interactive way.
- 60. As a user, I want to earn rewards for solving maze challenges using if-else statements and loops, so I feel motivated to apply these concepts effectively.
- 61. As a user, I want the looping challenges to scale in difficulty, like with different maze maps, so I can gradually improve my skills.
 - Learning Pointers:
- 62. As a user, I would like a quest involving keys, chests, and addresses to learn the basics of pointers and callback functions, so I can understand how to reference and use memory locations in a practical context.
- 63. As a user, I want an NPC to explain pointers with visual examples (e.g., diagrams showing memory addresses and values), so I can better grasp the concept.
- 64. As a user, I want to unlock pointer-related enchantments (e.g., dynamic item enhancements) after completing the quest, so I feel rewarded for mastering a complex concept.
- 65. As a user, I want to receive a specific achievement or rewards after learning pointers.
- Shop and Blacksmith Features
 - 66. As a user, I want to spend earned money in a shop to purchase blank functional cards, so I can customize my deck and strategy.

67. As a user, I want the blacksmith to enchant functional cards with achieved functions, so my cards can gain new abilities and become more versatile.
68. As a user, I want access to magic cards that provide permanent stamina boosts or extra coin rewards, so I can gain long-term advantages in the game.
69. As a user, I would like to upgrade my functional cards for better performance, so they are more effective during battles.
70. As a user, I want the shop to offer limited-time items (e.g., rare cards, consumables) that rotate daily or weekly, so I feel incentivized to check back regularly.
71. As a user, I would like to trade in unused cards or items for coins or shop credits, so I can make better use of my resources.

Non-functional Requirements

- Cross-Platform Compatibility
Our game will run seamlessly on compatible systems. The Unity frontend will dynamically adjust to various screen sizes and resolutions to ensure a consistent and immersive user experience across all platforms.

- Security
The backend, implemented in **Node.js**, will use **HTTPS** for secure data transmission. User data will be protected with **basic encryption** before storage in **MongoDB**. Only authenticated users can access their own data.

- Performance
The game will ensure the following performance benchmarks:
 - The **main menu** will load within **6 seconds** on devices meeting minimum hardware requirements.
 - **Gameplay interactions** (e.g., playing a card, opening a menu, or navigating the map) will respond within **3 seconds** to maintain a smooth user experience.
 - Background tasks, such as saving progress or updating inventory, will execute within **500 milliseconds** under normal load conditions.

- Maintainability

The system will be designed for easy maintenance and future expansion:

- **Frontend and backend separation** will allow independent updates and debugging, ensuring that new features can be added with minimal disruption to existing functionality.
- The backend will follow a **modular design**, with key features (e.g., authentication, inventory management, shop, and battles) implemented as independent modules or services.
- Usability:

Compared to other existing similar applications, the usability of *Bareturn0*; is designed to provide an intuitive and seamless gaming experience, focusing on a clean and visually appealing interface that balances functionality with aesthetics. From the moment players launch the game, they are greeted with a polished loading animation that can be skipped for convenience, ensuring a smooth entry into the main menu, where all essential functions such as account management, settings, and progress tracking are easily accessible. The game emphasizes clarity and simplicity, with minimal clutter, intuitive icons, and clear visual cues to guide players through every interaction. Real-time updates, such as health bars, stamina meters, and card pile states, keep players informed without overwhelming them, while features like customizable save slots, deck-building tools, and personalized profiles enhance user engagement and adaptability. The game reinforces motivation through immediate feedback, such as on-screen notifications for achievements and rewards, creating a satisfying sense of progression. By combining intuitive navigation, strategic customization, and visually engaging elements, *Bareturn0*; ensures an immersive and user-friendly experience that appeals to both new and experienced players.